

```
import psycopg2
import pandas as pd

# Connect to PostgreSQL
conn = psycopg2.connect(
    host="localhost",
    database="retail_analytics",
    user="postgres",
    password="Sneha2004"
)

print("Connection successful")
```

Connection successful

```
[16]: import datetime as dt

# Latest date in dataset
today = sales['invoice_date'].max()

# Create RFM table again
rfm = sales.groupby('customer_id').agg({
    'invoice_date': lambda x: (today - x.max()).days,
    'invoice_no': 'count',
    'total_amount': 'sum'
})

# Rename columns
rfm.columns = ['Recency', 'Frequency', 'Monetary']

rfm.head()
```

```
[16]:
```

	Recency	Frequency	Monetary
customer_id			
12346.0	325	2	0.00
12347.0	1	182	4310.00
12348.0	74	31	1797.24
12349.0	18	73	1757.55
12350.0	309	17	334.40

```
rfm['R_score'] = pd.qcut(rfm['Recency'], 5, labels=[5,4,3,2,1])
rfm['F_score'] = pd.qcut(rfm['Frequency'].rank(method='first'), 5, labels=[1,2,3,4,5])
rfm['M_score'] = pd.qcut(rfm['Monetary'], 5, labels=[1,2,3,4,5])

rfm.head()
```

	Recency	Frequency	Monetary	R_score	F_score	M_score
customer_id						
12346.0	325	2	0.00	1	1	1
12347.0	1	182	4310.00	5	5	5
12348.0	74	31	1797.24	2	3	4
12349.0	18	73	1757.55	4	4	4
12350.0	309	17	334.40	1	2	2

```
def segment_customer(row):

    if row['RFM_Score'] == '555':
        return 'Champions'

    elif row['R_score'] >= 4 and row['F_score'] >= 4:
        return 'Loyal Customers'

    elif row['R_score'] >= 3 and row['F_score'] >= 3:
        return 'Potential Loyalists'

    elif row['R_score'] <= 2 and row['F_score'] >= 3:
        return 'At Risk'

    else:
        return 'Hibernating'

rfm['Segment'] = rfm.apply(segment_customer, axis=1)

rfm.head()
```

	Recency	Frequency	Monetary	R_score	F_score	M_score	RFM_Score	Segment
customer_id								
12346.0	325	2	0.00	1	1	1	111	Hibernating
12347.0	1	182	4310.00	5	5	5	555	Champions
12348.0	74	31	1797.24	2	3	4	234	At Risk
12349.0	18	73	1757.55	4	4	4	444	Loyal Customers
12350.0	309	17	334.40	1	2	2	122	Hibernating

```
rfm['Segment'].value_counts()
```

Segment	
Hibernating	1749
Potential Loyalists	844
Loyal Customers	793
At Risk	669
Champions	317
Name: count, dtype: int64	